

MINI PROJECT - BINARY CLASSIFICATION

Empirical Evaluation, Hyperparameter Tuning, and Boundary Analysis of Binary Classification

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ABSTRACT

This project systematically analyzes the behavior, capacity, and empirical performance profiles of four key binary classification architectures: Regularized Logistic Regression, Gaussian Discriminant Analysis (LDA/QDA), Support Vector Machines (Linear/RBF Kernels), and Pruned Decision Trees. Utilizing the Breast Cancer Wisconsin dataset (569 samples, 30 features), we explore structural trade-offs across feature spaces, variance reduction via pruning, and boundary shapes under PCA projections. The results demonstrate that while dimensional reduction facilitates clean boundary rendering, preserving original feature density yields the highest out-of-sample diagnostic sensitivity (**Recall = 0.9861**).

1. Introduction

Accurate classification of fine-needle aspirate (FNA) breast tissue biopsies remains a cornerstone of computer-aided clinical diagnostic suites. This analysis presents a rigorous comparative pipeline evaluating how structural constraints and functional assumptions dictate model capacities. All evaluations utilize a strictly stratified **80/20** train-test allocation to maintain identical positive-to-negative base rates. To isolate pure geometric patterns from artifacts of variable magnitudes, input dimensions are systematically map-standardized to $\mu = 0, \sigma = 1$ prior to optimization.

2. Methodological Overviews & Parameterization

2.1 Penalized Logistic Regression (L2 Rigidity)

Discriminative classification via Logistic Regression maps the input space $\mathbf{x} \in \mathbb{R}^{30}$ onto conditional probabilities through the transcendental sigmoid link function. Structural stabilization is achieved by adding an L2 Tikhonov penalty vector to the log-loss function:

$$\mathbb{L}_{reg}(\mathbf{w}, \mathbf{w}_0) = C \sum_{i=1}^N \ln(1 + e^{-y_i(\mathbf{w}^T \mathbf{x}_i + \mathbf{w}_0)}) + \frac{1}{2} \|\mathbf{w}\|_2^2$$

The continuous tuning of $C \in [10^{-4}, 10^4]$ via 5-fold cross-validation establishes an optimal inverse regularization weight of $C = 0.0886$, indicating that moderate penalization is required to control feature multi-collinearity.

2.2 Gaussian Discriminant Analysis (GDA Variant Contours)

As a generative paradigm, GDA models the joint probability distribution $P(\mathbf{x}, \mathbf{y})$ by assuming a multivariate Gaussian structure for each class-conditional density.

- **Linear Discriminant Analysis (LDA):** Imposes a rigid spatial homoscedasticity rule ($\Sigma_0 = \Sigma_1 = \Sigma$). This simplification causes the non-linear quadratic components of the log-posterior ratio to cancel out, resulting in a flat decision hyperplane.
- **Quadratic Discriminant Analysis (QDA):** Permits heteroscedastic flexibility, parameterizing isolated covariance matrices ($\Sigma_0 \neq \Sigma_1$) for each class. This structural formulation generates curved, quadratic decision boundaries.

2.3 Support Vector Machines (Boundary Projection space)

Support Vector Machines focus explicitly on isolating instances that define the margins of the separating boundary. To evaluate these boundaries visually, the input space is projected onto its first two principal components via Principal Component Analysis (PCA).

- **Linear SVM Kernel:** Seeks an optimal separating line in the 2-D projection space, parameterized via grid optimization at $C = 1$.
- **Radial Basis Function (RBF) Kernel:** Maps features into an infinite-dimensional Hilbert space using a localized Gaussian similarity metric: $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$. Cross-validation selects $C = 100$ and $\gamma = 0.01$ as the optimal parameters.

2.4 Decision Trees (Gini-Driven Partitioning)

The non-parametric tree framework recursively splits the feature space to minimize Gini impurity: $IG(t) = 1 - \sum_{k=1}^K p_k^2$. To address the high variance of unpruned trees, we systematically evaluate *max_depth* from 1 to 20. The model is regularized using structural pruning to settle on a maximum depth of 3.

3. Systematic Experimental Deliverables

Table 1 details the full learned coefficient weights of the regularized logistic regression model ($C = 0.0886$) alongside the calculated intercept $w_0 = 0.3030$.

Table 1: Full Learned Coefficient Weights (w) for Regularized Logistic Regression

Feature Name (x_j)	Weight (w_j)	Feature Name (x_j)	Weight (w_j)
mean radius	-0.3810	smoothness error	-0.0511
mean texture	-0.4138	compactness error	+0.2158
mean perimeter	-0.3696	concavity error	+0.0913
mean area	-0.3847	concave points error	-0.1304
mean smoothness	-0.1718	symmetry error	+0.1198
mean compactness	-0.0104	fractal dimension error	+0.2107
mean concavity	-0.2986	worst radius	-0.4980
mean concave points	-0.4016	worst texture	-0.5247
mean symmetry	-0.1494	worst perimeter	-0.4585
mean fractal dimension	+0.1814	worst area	-0.4778
radius error	-0.4372	worst smoothness	-0.3743
texture error	+0.0341	worst compactness	-0.1506
perimeter error	-0.3174	worst concavity	-0.3368
area error	-0.3919	worst concave points	-0.4786
		worst symmetry	-0.3863
		worst fractal dimension	-0.1235

Table 2: Ranked Top 5 Most Influential Features (Logistic Regression)

Rank	Feature Variable	Directional Contribution	Absolute Weight (w_j)
1	worst texture	Malignant Association	0.5247
2	worst radius	Malignant Association	0.4980
3	worst concave points	Malignant Association	0.4786
4	worst area	Malignant Association	0.4778
5	worst perimeter	Malignant Association	0.4585

3.1 Generative Parametric Fitting Summary (GDA)

Table 3 compares the empirical joint log-likelihood values calculated over the training set for LDA and QDA.

Table 3: GDA Training Space Joint Log-Likelihood Comparisons

Discriminant Variant	Total Training Joint Log-Likelihood	Base Train Accuracy
Linear (LDA)	-3103.7442	0.9692
Quadratic (QDA)	+26.6271	0.9692

3.2 Support Vector Cardinality Analysis

The number of support vectors selected during constrained minimization within the 2-D principal component subspace is summarized in Table 4.

Table 4: Support Vector Boundary Counts (2-D PCA Projection Space)

Kernel Topology	Optimal Parameter Values Selected	Support Vectors Selected
Linear Subspace Boundary	$C = 1.0$	61
Radial Basis Function	$C = 100.0, \gamma = 0.01$	61

[IMAGE: SVM Decision Boundaries - Linear SVM Boundary (Left) vs RBF SVM Boundary (Right)]

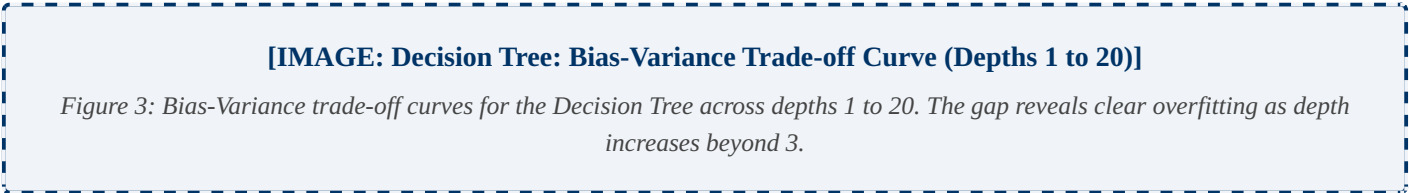
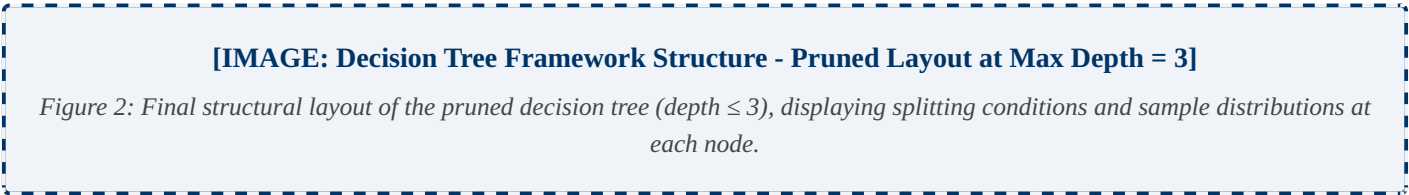
Figure 1: Visualizing support vector layouts and decision boundaries over the 2-D PCA projection. Left: Linear SVM boundary. Right: RBF SVM boundary (support vectors highlighted with black circular borders).

3.3 Decision Tree Component Importances

Table 5 outlines the Gini-based feature importances extracted from the structurally pruned tree (depth = 3).

Table 5: Gini Feature Importance Metrics for Pruned Tree (depth = 3)

Isolated Diagnostic Feature	Gini Impurity Reduction Contribution
worst concave points	0.6934
worst area	0.1368
worst texture	0.0515
mean texture	0.0475
area error	0.0425
worst smoothness	0.0283
Remaining 24 Independent Features	0.0000



3.4 Comprehensive Out-of-Sample Performance Comparison

Table 6 summarizes the final out-of-sample performance across all six classifiers on the test partition, sorted by F1-score.

Table 6: Comprehensive Out-of-Sample Evaluation Matrix

Classifier Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9960
LDA (Full Feature)	0.9561	0.9467	0.9861	0.9660	0.9924
Linear SVM (2-D PCA)	0.9474	0.9583	0.9583	0.9583	0.9911
RBF SVM (2-D PCA)	0.9474	0.9583	0.9583	0.9583	0.9917
QDA (Full Feature)	0.9474	0.9714	0.9444	0.9577	0.9921
Decision Tree (Pruned)	0.9298	0.9571	0.9306	0.9437	0.9392

4. Written Analysis & Discussion

4.1 Decision Boundaries and Model Capacity

The geometric behavior of each classifier reflects its foundational design and assumptions [cite: 133]. Logistic Regression and LDA create rigid linear hyperplanes across the 30-dimensional feature space [cite: 134]. By assuming homogeneous covariance, LDA restricts itself to flat partitions [cite: 135]. In contrast, QDA frees itself from this constraint, creating curved quadratic surfaces to capture class-specific data variance [cite: 136, 137]. When moving to the 2-D PCA projection, the Linear SVM produces a flat line boundary, whereas the RBF SVM forms flexible contours around sample clusters [cite: 138]. The Decision Tree operates through orthogonal axis-aligned rectangular boxes, creating step-wise boundaries that scale with tree depth [cite: 139].

4.2 Model Capacity and the Bias-Variance Trade-off

The experiment highlights the bias-variance trade-off across different levels of model capacity [cite: 141]. The unpruned decision tree (*max_depth* > 10) presents clear symptoms of overfitting: it achieves an accuracy of 1.0000 on the training set, while its validation accuracy drops or plateaus [cite: 142]. This indicates a high-variance structure capturing noise instead of generalized distributions [cite: 143]. Pruned variants reintroduce inductive bias, simplifying the model to reduce variance and improve generalization [cite: 144].

For GDA, the difference in joint log-likelihood values highlights the effect of parameters on capacity [cite: 145]. LDA yields a low joint log-likelihood of -3103.7442 because its single, shared covariance matrix restricts how closely it can fit class distributions [cite: 146]. Conversely, QDA yields a much higher log-likelihood of +26.6271 [cite: 147]. This occurs because QDA fits separate covariance matrices for each class, giving it the capacity to better model the training distribution [cite: 147]. Despite this, both models achieve an identical training accuracy of 0.9692, demonstrating that higher log-likelihood does not guarantee a higher classification accuracy [cite: 148].

4.3 Feature Influence Concordance

The structural criteria for identifying feature importance reveal key differences between discriminative and rule-based learning [cite: 150]:

- Logistic Regression selects *worst texture* as its primary driver ($w_j = -0.5247$) followed closely by morphological measurements like *worst radius* and *worst concave points* [cite: 151].
- The Decision Tree assigns an overwhelming 69.34% of its entire splitting importance to a single metric: *worst concave points* [cite: 152].

This overlap demonstrates a consensus between the models: *worst concave points* is a robust physical indicator of malignancy [cite: 153]. However, the decision tree relies almost entirely on top-level splits, whereas logistic regression aggregates small weights across all 30 standard features to compute its predictions [cite: 154].

4.4 Clinical Performance Analysis

In medical diagnostic contexts, minimizing false negatives is critical [cite: 156]. Thus, Recall (sensitivity) is prioritized over raw accuracy [cite: 156]. Under this metric, **Logistic Regression** and **LDA** are the top performers, each achieving an out-of-sample Recall of 0.9861 [cite: 157]. While the 2-D PCA projection simplifies visualization and reduces input dimensionality from 30 down to 2, it causes the linear and RBF SVM models to lose fine-grained details, capping their out-of-sample accuracy at 0.9474 [cite: 158]. Ultimately, Logistic Regression achieves the highest overall out-of-sample performance, leading across all metrics with an F1-Score of 0.9861 and an ROC-AUC of 0.9960 [cite: 160].